Blake Grudzien

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Experience

Level Home May 2025 – September 2025
Software Engineer Intern | *Case study* *Redwood City, CA*

- Built the Go service behind a rebuilt device telemetry pipeline on AWS (Lambda, SQS, S3, Athena), cutting data loss from over 30% to under 1% across millions of events from smart locks and video doorbells, and presented the architecture to 40+ people across the company
- Replaced silent deletion of unparseable files with a dead-letter queue, so the firmware team could ship a parser fix and clear the entire backlog with one redrive instead of a cross-team coordination cycle
- Chose SQS over EventBridge after finding fan-out irrelevant for a single-consumer path, and kept the native DLQ over a custom one, trading per-error retry control for zero custom code and one-command reprocessing
- Migrated telemetry storage from row-based JSON to Parquet via Apache Arrow so Athena scans only queried columns, choosing Arrow over DuckDB, whose CGo requirement would have forced changes across the service
- Built the service's test suite and CI pipeline, covering the full path from a raw telemetry file to queryable Parquet output
- Prototyped a Slack bot alerting on every parse failure, then narrowed it to a daily digest of the queue after the firmware team judged per-failure alerts to be noise

Brooksee June 2024 – October 2024
Software Engineer Intern | *Case study* *Salt Lake City, UT (Hybrid)*

- Built a Garmin Connect IQ data field (Monkey C) and Kotlin Android app to stream live pace, distance, placement, and gun-time gaps to runners' watches for races of 5,000–10,000 participants, tested against a real-time race simulation
- Computed the gap to the runner ahead, which the data never stored, by comparing timestamped crossings at each timing point, and delivered it through a Firestore listener on Android bridged to the watch over Bluetooth
- Derived each runner's elapsed time from the race's absolute start timestamp rather than a local timer, so runners crossing the line at irregular intervals after the gun would still see correct official time despite device clock drift

Education

Emory University Atlanta, GA
B.S. Computer Science, Minor in Mathematics | *GPA: 3.65/4.00* *May 2026*

Projects

Chess Scholar (Live | Github | Case study) | ***Python, Claude API, PostgreSQL/pgvector*** July 2026 – Present

- Built a hybrid retrieval system over 127,435 games and 10.2M moves (1610–2023), routing questions across deterministic SQL, pgvector search over 468,862 embedded annotation chunks, and pooled Stockfish subprocesses through Claude's native tool-calling rather than LangChain or a hand-rolled router
- Required engine grounding before any model claim about move quality, constraining the LLM to attribute answers to retrieved stats, prose, or evaluations rather than its own judgment
- Trained a gradient-boosted classifier (0.887 cross-validated ROC-AUC) to filter scraped study material before surfacing it, and covered the system with 250+ tests running against Python 3.11 and 3.12 in CI
- Wrapped three paid APIs and PostgreSQL in explicit error handling, with an 8 req/min session cap to stop runaway usage

Bird Strike Analysis (Live | Github) | ***SvelteKit, D3.js, TypeScript*** Feb 2026 – May 2026

- Built an interactive data-driven article analyzing 300k+ FAA bird strike records, featuring animated geographic maps, linked charts, and a species heatmap, deployed via GitHub Actions

CHOA Chatbot (Github) | ***Python, React, MongoDB*** Jan 2024 – May 2024

- Built an AI-powered chatbot for Children's Healthcare of Atlanta in a 7-person team, training resident doctors on pediatric airway management protocols for respiratory emergencies

Technical Skills

Languages: Python, Go, SQL, TypeScript, JavaScript, Kotlin, Java, R
Frameworks & Libraries: Flask, React, SvelteKit, Streamlit, Scikit-Learn, pandas, NumPy, pytest, D3.js
Cloud & Data: AWS (Lambda, SQS, S3, Athena, Kinesis), Parquet, PostgreSQL/pgvector, Firestore, Cloudflare Workers
Tools & Other: Git, GitHub Actions, Docker, Linux, REST APIs, RAG, vector embeddings, LLM tool-calling

Leadership

Vice President Oct 2023 – Sep 2024
Alpha Tau Omega Fraternity, Emory University (100+ members) *Atlanta, GA*

- Led chapter to 3rd highest GPA of 136 national ATO chapters; headed the disciplinary process, mediating conflicts

Additional

Chess: 2209 chess.com elo (99.9 percentile) | **Languages**: Intermediate Spanish | **Interests**: Tennis, Rock Climbing, Soccer